VISUALIZATION

Course Outline

Course Objective

- Learn the principles involved in designing effective information visualizations.
- Understand the wide variety of information visualizations and know what visualizations are appropriate for various types of data and for different goals.
- Understand how to design and implement information visualizations.
- Know how information visualizations use dynamic interaction methods to help users understand data.
- Learn to apply an understanding of human perceptual and cognitive capabilities to the design of information visualizations.
- Develop skills in critiquing different visualization techniques in the context of user goals and objectives.
- Gain knowledge of some popular visualization tools and libraries.
- Understand the concept of visualization grammars.
- Introduction to Scientific Visualization.
- Know how AI techniques are used in Visualization tasks
- Know how Visual Analytics can help in interpreting Deep Neural Networks.

Syllabus

- Introduction
- Fundamentals of Information Visualization
  - Data
  - Visual Encoding
  - Interaction techniques
  - Graphical Excellence
- Visualization of common data sets
  - Tables
  - Trees
  - Graphs
  - Text
- Implementation
  - Visualization tools and libraries
  - Visualization grammars
- Introduction to Scientific Visualization
- Visualization & AI
  - Visualization analytics for Explainable AI
  - AI for Visualization generation and understanding

Why take this course?

- Industry:
  - Visualization is a key component of Data Science
  - Visualization tools are used extensively
- Research:
  - Visualization and Visual Analytics are active areas of Research
  - Augments research in other areas like Deep Learning, Medical Informatics, etc.

Instructor

- Sougata Mukherjea
  - PhD, Georgia Institute of Technology
  - Email: sougatam@iitd.ac.in
  - Mobile/WhatsApp: 9810886820

- TAs
  - Vaibhav Kagathara

Grading

Type of Assessment	Percentage
Assignments	30%
Project	25%
Minor	20%
Major	25%

For Audit Pass minimum C grade

Attendance Policy

- Students with attendance less than 50% can have their final grade reduced (Only for full-time students)

Logistics

- Tools
  - Moodle
- Slot: AB   Tu Fr 2-3:20pm
- Room: LH413.2
- Prerequisites
  - Programming languages (Python)
  - Data structures

Text Book

- Information Visualization, Sougata Mukherjea, BPB Publishers, 2025
  - Amazon